Tailong Xiao

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Research Interests

Quantum Artificial Intelligence (QAI); Machine Learning; Intelligent Quantum Sensing; AI-enhanced Optical Computing and Imaging.

Academic Appointments

Oct. 2023–Present	**Assistant Professor**, SJTU, Shanghai, China.

Education

2018-06	**B.E.**, Communication Engineering, Central South University, China.
2023-06	**Ph.D.**, Information and Communication Engineering, SJTU, China.

Selected Publications

1. Xu H, Deng X, Zheng Z, **Xiao T.***, et al. Noise-resilient Heisenberg-limited quantum sensing via indefinite-causal-order error correction. *Physical Review Letters*, 2026, 137(2): 020801.

2. Zhu, Y., **Xiao, T.***, Zeng, G. et al. Controlling unknown quantum states via data-driven state representations. *npj Quantum Information*, (2026). https://doi.org/10.1038/ s41534-026-01269-0.

3. Li, Y., Wang, J. **Xiao, T.***, et al. Real-time imaging through moving scattering media enabled by fixed optical modulations. *Photonics Research*, 2026, 14(4): 1230.

4. Xu H, **Xiao T.***, Huang J, et al. Toward Heisenberg Limit without Critical Slowing Down via Quantum Reinforcement Learning. *Physical Review Letters*, 2025, 134(12): 120803.

5. **Xiao, T.**, Zhai, X., Huang, J. et al. Quantum deep generative prior with programmable quantum circuits. *Communications Physics*, 2024, 7: 276. DOI: 10.1038/s42005-024-01765-9.

6. **Xiao, T.**, Zhai, X., Wu, X. et al. Practical advantage of quantum machine learning in ghost imaging. *Communications Physics*, 2023, 6: 171. DOI: 10.1038/s42005-023-01290-1.

7. **Xiao, T.**, Fan, J. & Zeng, G. Parameter estimation in quantum sensing based on deep reinforcement learning. *npj Quantum Information*, 8, 2 (2022). DOI: 10.1038/s41534-021-00513-z.

8. **Xiao, T.**, Huang, J., Li, H., Fan, J., Zeng, G. Intelligent certification for quantum simulators via machine learning. *npj Quantum Information*, 8, 138 (2022). DOI: 10.1038/s41534-022-00649-6.

* indicates corresponding author. Full list: Google Scholar.

Teaching

- **Engineering Practice and Technological Innovation II (EE0503)**, Spring 2025. Topic: *An Introduction to Quantum Artificial Intelligence* (Undergraduate).

Research Grants & Projects

- **National Key R&D Program of China** (Dec. 2025–Nov. 2030).
- **Shanghai Municipal Science and Technology Commission** (Oct. 2025–May. 2026).
- **NSFC Young Scientists Fund** (2025–2027).
- **Lenovo–SJTU Joint Lab Project** (2024–2025).
- **SJTU Young Scholars Project** (2023–2026).
- **CCF–Boson Quantum Innovation Fund** (2025.03–2025.08).
- **National Key Laboratory Open Research Program** (2025–2026).

Academic Service

Professional Roles

- Executive Committee Member, CCF Technical Committee on Quantum Computing.

Editorial Work

- Guest Editor, Special Issue "Quantum Computing and Quantum Information Processing", *Entropy* (MDPI).
  Special issue link.
- Editorial Board Member, *Scientific Reports*.

Reviewer

- Physical Review Letters / Applied / Research / A; PRX Quantum.
- New Journal of Physics; Quantum Science and Technology; Journal of Physics A; Machine Learning: Science and Technology; European Journal of Physics.
- npj Quantum Information; Scientific Data; Scientific Reports; Quantum Information Processing.
- Laser & Photonics Reviews, ACS Photonics, Optica Quantum; Optics Express; Optics Letters.

Honors & Awards

- CCF Sinan Cup, National Second Prize, 2021.
- CCF Sinan Cup, National First Prize, 2022.
- CCF Outstanding Doctoral Dissertation in Quantum Computing, 2024.
- Outstanding Achievement Case, the 4th CCF Conference on Quantum Computing (CQCC 2025)